

Total No. of Questions : 8]

SEAT No. :

P800

[Total No. of Pages : 2

[5870] - 1118

T.E. (Chemical)

CHEMICAL REACTION ENGINEERING -I
(2019 Pattern) (Semester -II) (309348)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Answer Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figures to the right side indicate full marks.
- 4) Assume suitable data, if necessary.

Q1) a) Derive performance equation for Steady state batch and show graphical representation. **[12]**

- b) A homogeneous liquid phase second order reaction takes place in MFR with 50% conversion. Find conversion if original reactor is replaced by another MFR having volume 6 times that of the original reactor-all else remain unchanged. **[6]**

OR

Q2) a) Derive performance equation for steady state Plug Flow Reactor and show graphical representation. **[12]**

- b) Explain the terms holding time and space time in detail. **[6]**

Q3) a) Explain series, parallel, and independent reactions. With suitable examples **[12]**

- b) Derive the relationship for C_{Rf} in terms of ψ for CSTR in case of parallel reaction. **[5]**

OR

P.T.O.

Q4) In CSTR, reactant A produces Products R and S by parallel reaction, $r_R = 0.4C_A^2$ and $r_S = 2C_A$. A feed with $C_{A0} = 40$ enters a MFR and mixture of A, R and S leaves the reactor. Find C_R , C_S and τ for 90% conversion in CSTR. [17]

- Q5)** a) Explain effect of temperature on heat of reaction with equations. [12]
b) Discuss the Van't Hoff isotherm equation. [6]

OR

- Q6)** a) Explain non-adiabatic operation in detail. [12]
b) Explain the characteristics of chemical equilibrium. [6]

- Q7)** a) Explain tank in series model in detail. [9]
b) Explain the concept of residence time distribution in detail. [8]

OR

- Q8)** What do you mean by E, F and C curves? Explain relationship between E, f and C curves. [17]

